Computer Science

Senior Capstone Report - Fall 2024

Kiwi+: A Context-Aware Educational Chatbot Enhanced by Knowledge Retrieval and Multi-Modality

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Declaration

I declare that this senior capstone was composed entirely by myself with the guidance of my advisor, and that it has not been submitted, in whole or in part, to any other application for a degree. Except where it is acknowledged through reference or citation, the work presented in this capstone is entirely my own.

Preface

The Kiwi+ project was inspired by the experiences of both authors as undergrad students in Computer Science during the rise of Large Language Models (LLMs) as educational tools. Both authors, having worked together in various classes, noticed that traditional AI systems often fail to provide context-specific explanations and instead offer direct answers, limiting effective learning. Kiwi+ seeks to address this gap by leveraging techniques like Retrieval-Augmented Generation (RAG), the system fosters deeper engagement and better comprehension, enabling more context-aware and enhances the learning experience by integrating audio and video modalities, allowing for more accessible learning experiences. This project is tailored for students, educators, and institutions looking for more personalized, multimodal learning tools. It is particularly beneficial for students with diverse learning preferences or accessibility needs, offering content in a variety of formats.

Acknowledgements

We extend our heartfelt thanks to Prof. Hongyi Wen and Multimodal Adaptive Personalization Systems Research group for their invaluable guidance and support throughout this project. Their mentorship and insightful feedback were instrumental in shaping our work and helping us overcome challenges. We are deeply grateful for their encouragement and for sharing their expertise with us.

Abstract

The rapid advancement of large language models and large vision-language models has revolutionized the field of artificial intelligence, especially in education. Kiwi+, a context-aware chatbot, aims to address this gap by leveraging techniques like Retrieval-Augmented Generation to enhance AI-driven tools by integrating multiple data modalities (text,image,audio, and video) with advanced knowledge retrieval mechanisms. In this study, we evaluate the performance of a multimodal RAG-based chatbot using a comprehensive experimental protocol, focusing on retrieval quality, factuality, and faithfulness. Our findings indicate that, while the retrieval mechanism effectively identifies relevant context, the response generation sometimes deviates from the expected alignment, particularly in cases with limited data.

Keywords

Capstone, Computer science, NYU Shanghai, Knowledge retrieval, Natural language processing, AI in education, Human-computer interaction, Personalized learning, Multimodality, LLMs, MLOps

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1 Introduction

The rapid advancement of large language models (LLMs) and large vision-language models (LVLMs) has revolutionized the field of artificial intelligence (AI), especially in education. AI-based educational tools have significantly benefited from LLMs, which offer natural conversation abilities, instructional support, and adaptability to multimodal inputs such as images, text, code, and voice. These capabilities have paved the way for more advanced, context-aware educational chatbots.

Building on the success of prior systems like Kiwi-ICP and AI-driven platforms like Khan Academy AI, this report introduces Kiwi+, a context-aware chatbot specifically designed to enhance computer science education. Kiwi+ integrates Retrieval-Augmented Generation (RAG) with advanced Knowledge Retrieval (KR) and Knowledge Learning (KL) techniques to deliver more personalized and contextually aware learning experiences. Unlike traditional AI educational tools that rely on static, text-based interactions, Kiwi+ dynamically retrieves multimodal content—including text, youtube videos, code, images, and lecture transcripts—in response to student queries, creating a more interactive and adaptive learning environment. This research aims to transform education through Kiwi+, multimodal chatbot solution that reduces the need for costly model fine-tuning while providing personalized, real-time support to learners.

2 Related Work

Multimodal retrieval-augmented generation (RAG) systems have gained significant attention for their ability to enhance AI-driven tools by combining multiple data modalities with advanced knowledge retrieval mechanisms. One such system, Kiwi-ICP, has demonstrated remarkable potential by enabling interactions with pre-uploaded RAG databases, including assignments, lecture slides, and exams. This functionality set a strong foundation for educational AI systems, offering a robust framework for accessing and utilizing structured academic content. However, Kiwi-ICP lacked two critical features: support for user-uploaded content that can be dynamically integrated into a relational database and the inclusion of audio and video modalities. These limitations directly inspired our work, which extends Kiwi-ICP to address these gaps.

Our system introduces a dynamic content integration mechanism that allows users to upload new data, which is immediately processed and stored in a relational database. This enhancement significantly increases the system's adaptability and utility, enabling it to support a broader range of educational scenarios. Unlike the static database reliance of Kiwi-ICP, our approach ensures that the chatbot can evolve in real-time with the addition of new resources, aligning with the growing demands of adaptive and personalized learning environments.

Additionally, the inclusion of audio and video input modalities represents a significant leap forward. While prior systems like QWEN2-Audio [1] and Moshi [2] explored the potential of speech-to-text and real-time dialogue systems, their scope was largely limited to conversational AI without robust retrieval capabilities. Our work integrates these modalities into a unified multimodal RAG framework, allowing students to upload lecture recordings or video presentations for analysis and retrieval. This multimodal capability bridges a crucial gap, offering richer interactions and enabling students to engage with content in ways that mimic real-world educational settings.

Building on related innovations, such as the multimodal question-answering capabilities of MuRAG [3] and the personalized support offered by LLM-based tools in studies like Kazemitabaar et al. [4], our system combines these elements into a cohesive, domain-specific tool. While MuRAG excels at integrating text and image modalities, it does not account for the dynamic and evolving nature of educational content. Similarly, while Kazemitabaar et al. focus on enhancing comprehension for novice programmers, they lack multimodal support and the ability to retrieve resources from complex databases. Our chatbot addresses both limitations by providing real-time retrieval and adaptive interactions across text, image, audio, and video modalities.

Moreover, our work expands on knowledge graph-based retrieval strategies discussed by Abu-Rasheed et al. [5] by enabling not only static knowledge representation but also the dynamic enrichment of the knowledge base. This continuous updating ensures that the chatbot remains current with newly uploaded or generated content, a feature absent in many existing systems.

By integrating unstructured inputs like lecture recordings, video explanations, and live interactions, our system also builds on the multimodal frameworks introduced in VizChat [6], which focus on structured visual narratives. Our approach extends this work to handle unstructured and diverse data types, increasing the system's versatility for varied educational applications.

In summary, while Kiwi-ICP provided a solid foundation for RAG-based educational chatbots, our work significantly extends its capabilities by incorporating user-uploaded data, relational database integration, and multimodal inputs. These advancements collectively position our system as a next-generation tool for personalized and immersive learning, addressing the evolving needs of students and educators alike.

3 Solution

3.1 Preprocessing

To facilitate the integration of audio and video modalities, we designed a comprehensive preprocessing pipeline that transforms raw video data into structured and usable components. The pipeline, as illustrated in Fig. 1, is optimized to handle and process YouTube video URLs, ensuring efficient data transformation and preparation. The key stages of the pipeline are as follows:

Initially, the pipeline automates the acquisition of video content by downloading the specified media directly from YouTube links, which streamlines the data acquisition process and ensures consistency. The downloaded video content is then systematically segmented into audio and video components, generating separate audio files in `.wav` format and video files in `.mp4` format. This separation facilitates independent and targeted processing of each modality.

To extract valuable information from the video content, the pipeline incorporates advanced scene detection and optical character recognition (OCR) techniques. These methods are employed to detect and extract slide images from video frames and perform text extraction. The extracted slides are systematically paired with metadata, including timestamps and image identifiers, which are stored in a `JSON` format. This metadata serves as a foundational element for subsequent encoding and retrieval processes, ensuring temporal alignment and contextual relevance of the extracted content.

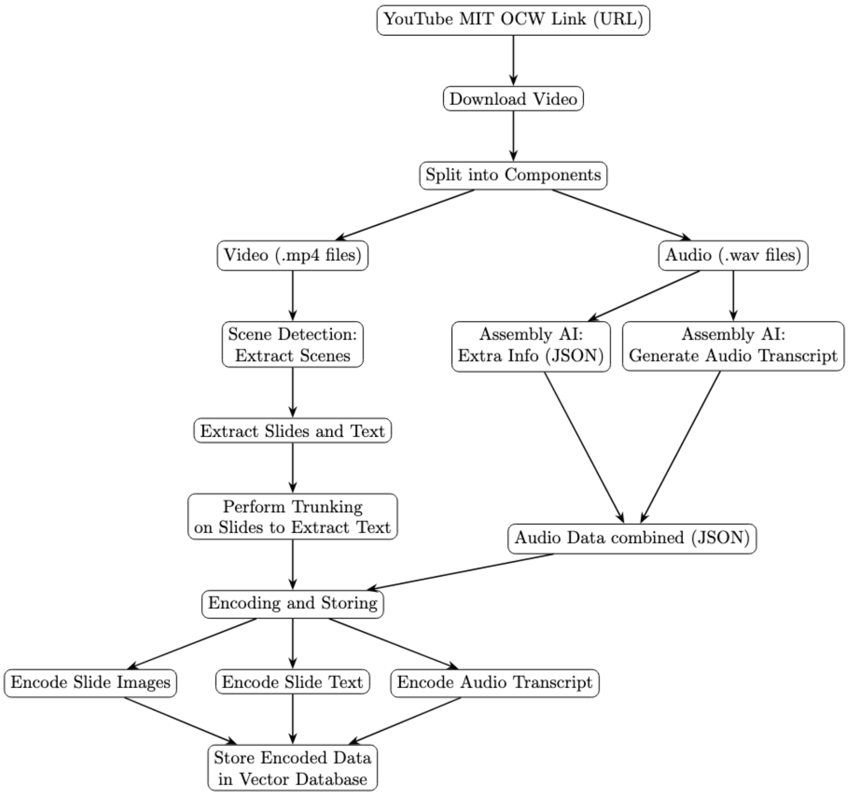

Figure 1: Preprocessing Flowchart for Video Encoding and Embeddings

3.2 Assembly AI Integration

The preprocessing pipeline is further enhanced by integrating Assembly AI to process and extract insights from the audio component of the data. As depicted in Fig. 1, the integration enables the generation of high-quality transcriptions and metadata, which are crucial for building a multimodal dataset.

The audio data is processed through a Assembly AI's transcription feature that converts audio signals into accurate, timestamped textual transcriptions with each speaker identified and noise removed. These transcriptions are further parsed into *gpt-4o mini* to get enriched summaries of each transcription chunk, providing a concise overview of the audio content. The enriched transcripts enhance the interpretability of the data, particularly in educational and analytical settings.

The structured data from the transcription process, along with relevant metadata, is organized into a `JSON` format. This structured representation is aligned with the slide-level data extracted earlier, ensuring temporal and contextual coherence across modalities. The `JSON` format encapsulates multimodal features such as timestamps, slide identifiers, and textual content, serving as a backbone for constructing multimodal vector representations.

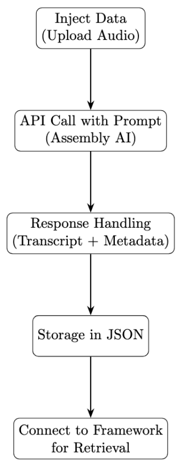

Figure 2: Assembly AI Pipeline for Audio Processing and Metadata Extraction

3.3 RAG Pipeline

The pipeline begins by processing the input data. If the input is video (e.g., a Youtube Video), as mentioned above, the audio of the youtube video is transcribed into text using the AssemblyAI API. Then, the text from the audio is broken into 45-second chunks to ensure manageable sizes. Each chunk is then summarized using *gpt-4o mini*, creating concise yet meaningful representations of the content. From these summaries, vector embeddings are generated. These embeddings numerically represent the key features of the text and are stored in ChromaDB, a specialized database for handling high-dimensional vectors. The embeddings are stored in ChromaDB for future retrieval.

When a user makes a query—whether by submitting text or providing audio—it is processed to ensure consistency. For audio queries, transcription and stop-word removal are performed, while text queries skip directly to cleaning. The cleaned query is then converted into an embedding, which serves as a vector representation of the user's request. This embedding is matched against the stored embeddings in ChromaDB to identify the top K most relevant entries.

In our system, we utilized cosine similarity as a measure to quantify the similarity between two vectors stored in ChromaDB. The cosine similarity between two vectors $\mathbf{t}$ and $\mathbf{e}$ is calculated using the formula:

$$\cos(t, e) = \frac{\mathbf{t} \cdot \mathbf{e}}{\|\mathbf{t}\|\|\mathbf{e}\|} = \frac{\sum_{i=1}^{n} t_i e_i}{\sqrt{\sum_{i=1}^{n} (t_i)^2} \sqrt{\sum_{i=1}^{n} (e_i)^2}}$$

where t_i and e_i are the components of vectors $\mathbf{t}$ and $\mathbf{e}$ respectively. $\sum_{i=1}^{n} t_i e_i$ denotes the dot product of vectors $\mathbf{t}$ and $\mathbf{e}$. $\sqrt{\sum_{i=1}^{n}(t_i)^2}$ and $\sqrt{\sum_{i=1}^{n}(e_i)^2}$ represent the magnitudes (Euclidean norms) of vectors $\mathbf{t}$ and $\mathbf{e}$ respectively. This formula computes the cosine of the angle between the two vectors, providing a value between -1 and 1, where 1 indicates perfect similarity, 0 indicates no similarity, and -1 indicates perfect dissimilarity. By comparing the cosine similarity values between vectors, we can effectively identify the most similar information in our dataset.

These retrieved entries are fed into a language model (LLM) to craft a comprehensive and contextually appropriate response for the user. This approach ensures accurate, efficient, and personalized results based on the input mode and stored data.

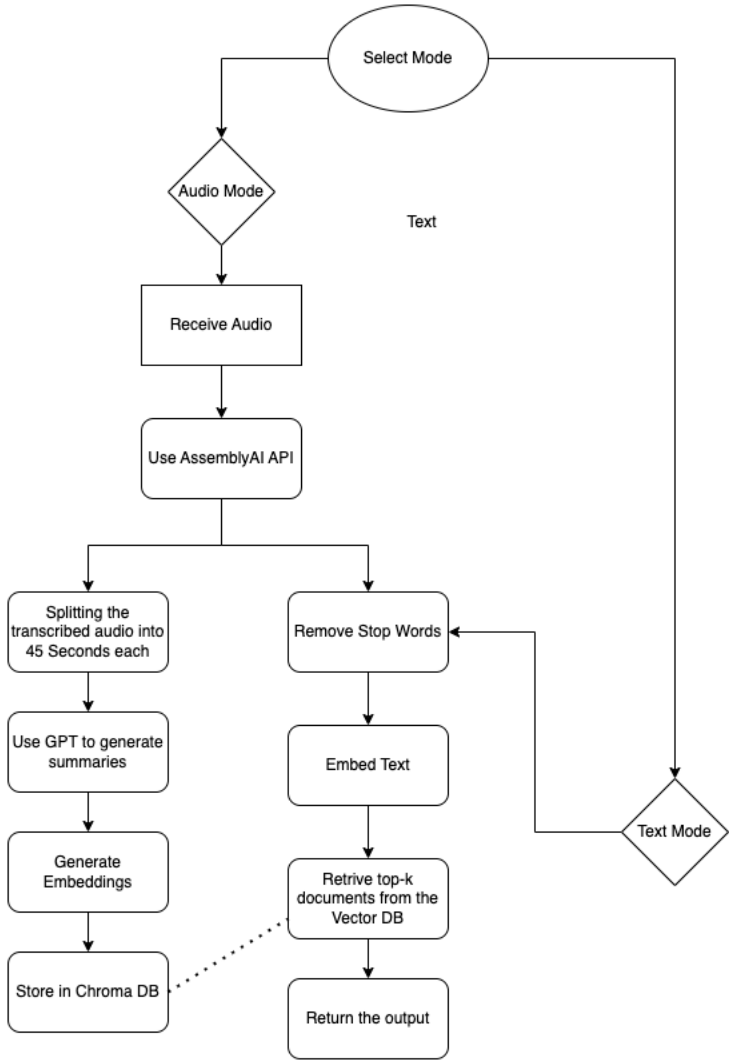

Figure 3: RAG Pipeline

3.4 Output

Once the RAG pipeline processes the query and feeds the relevant data into the LLM, the model generates a strategic response that is both contextually aware and tailored to the user's needs. The response is delivered in one of two modes: text or audio, depending on the user's initial selection. In audio mode, the output feels conversational and human-like, offering instantaneous, natural voice responses. This mode enhances interactivity and creates an immersive experience for the user. In text mode, the response functions like a chatbot, delivering clear and concise information. A standout feature of this system is its ability to extract context from videos quickly, summarize the relevant sections, and present the information interactively. This capability allows users to engage deeply with the content, learning efficiently and effectively. By seamlessly integrating these modalities, the system provides a robust and versatile tool for personalized learning and interaction.

4 Evaluation Metrics

4.1 Experimentation protocol

Experimental Protocol

In this study, we evaluate the performance of a Retrieval-Augmented Generation (RAG) system using a comprehensive experimental protocol. The evaluation employs three distinct metrics from the RAGAS framework [7] to assess retrieval quality, factuality, and faithfulness of the system. The evaluation is conducted on a set of queries, where the system retrieves responses that are subsequently analyzed across the chosen metrics. The results from each metric provide a holistic view of the system's performance, ensuring that the RAG system not only retrieves relevant information but also presents it in a coherent and accurate manner.

The experiment was conducted using Google's educational video "Intro to LLMs" [8], which is 15 minutes long. This duration provides sufficient content depth for building a comprehensive RAG model while maintaining a manageable experimental cost.

Five context-specific queries were posed for testing, each designed to elicit either a context-specific or generalized answer. This serves as a useful benchmark to determine whether the system is able to generate responses tailored to the specific context or if it defaults to more generalized answers.

- Query 1: How can LLMs be compared to the analogy of training a dog through commands?

- Query 2: How are large language models (LLMs) related to deep learning?
- Query 3: What makes large language models versatile?
- Query 4: What system does Palm use for multi-task training?
- Query 5: What features make Google's Palm API user-friendly for developers?

4.2 RAGAS (Retrieval-Augmented Generation Assessment Score)

This section summarizes the evaluation of the RAG system using RAGAS [7]. These tools assess the system's performance across three key dimensions: Retrieval Quality, Factuality, Faithfulness. The evaluation was conducted on a set of predefined queries and assessed via the following metrics and equations.

Retrieval Quality

Retrieval Quality refers to Answer Semantic Similarity which evaluates the alignment between a generated answer and the ground truth, with scores ranging from 0 to 1. A higher score indicates better similarity. This metric is computed using a cross-encoder model to assess response quality.

- Step 1: Vectorize the ground truth answer using the specified embedding model.
- Step 2: Vectorize the generated answer using the same embedding model.
- Step 3: Compute the cosine similarity between the two vectors.

Factuality

Factual Correctness evaluates the accuracy of a generated response by comparing it to the reference. The metric uses precision, recall, and F1 score to quantify factual overlap, with higher scores indicating better performance. It employs an LLM to break down both the response and reference into claims, using natural language inference to assess factual alignment.

$$\mathrm{F1} = \frac{2 \times Precision \times Recall}{Precision + Recall}$$

Faithfulness

The Faithfulness metric measures the consistency of a generated answer with the given context. It is calculated by comparing claims in the answer to the context, with scores scaled from 0 to 1. A higher score indicates better alignment, with the answer being faithful if all claims can be inferred from the context.

$$\text{Faithfulness score} = \frac{|\text{Number of claims in the generated answer that can be inferred from given context}|}{|\text{Total number of claims in the generated answer}|}$$

4.3 Results from RAGAS

Query	Retrieval Quality	Factuality	Faithfulness
Query 1	0.93	0.33	0.56
Query 2	0.91	0.09	0.18
Query 3	0.94	0.43	0.56
Query 4	0.95	0.57	0.5
Query 5	0.84	0.00	0.18

Table 1: Evaluation scores for each query from various metrics.

5 Discussion

Our project was successfully able to work on creating a streamlined pipeline for adding multi-modality in our system, and to allow users to interact with YouTube videos and audio modalities, in addition to text. We were successful in extending in the current kiwi system in this regard. In terms of the evaluation we have gotten a successful result on the retrieval quality in terms of semantic similarity, however the challenge on not getting a great factuality and faithfulness score, specially on Query 2 and 5 is worth mentioning and analyzing further.s

Query 1, which compares training large language models (LLMs) to training a dog through commands, achieved the highest retrieval quality (0.93) and decent scores for factuality (0.33) and faithfulness (0.56). This indicates that the system effectively retrieved relevant contexts but only moderately ensured factual alignment and coherence with the provided reference. Query 3, which explored what makes LLMs versatile, demonstrated improved overall performance, with retrieval quality at 0.94, factuality at 0.43, and faithfulness at 0.56, reflecting a more balanced alignment between retrieved contexts and the generated response. Query 4 achieved the highest retrieval quality (0.95) and showed relatively strong factuality (0.57) compared to other queries, though faithfulness slightly dropped (0.5). This suggests that the system effectively retrieved and processed relevant contexts, maintaining factual consistency to a reasonable degree.

In contrast, Query 2, focused on the relationship between LLMs and deep learning, also showed high retrieval quality (0.91) but fell significantly short in factuality (0.09) and faithfulness (0.18), suggesting that while the retrieval system identified related documents, it struggled to maintain a factual and coherent narrative. Upon reviewing the response in detail, it becomes clear that while it generally captures the relationship between LLMs and deep learning, it includes some inaccuracies. The mention of "up to petabytes" of training data is an exaggeration, as not all LLMs use such vast datasets, and the claim that LLMs have "billions of parameters" overlooks models with hundreds of billions of parameters. Additionally, the response simplifies the discussion of parameters, failing to reflect the full detail found in the retrieved contexts, which impacts its faithfulness. These discrepancies explain the low scores in factuality (0.09) and faithfulness (0.18), despite the high retrieval quality (0.91). On the other hand, Query 5 recorded the lowest factuality score (0.00) and very poor faithfulness (0.18), despite a high retrieval quality of 0.84. This indicates a significant gap between retrieved information and the generated response's alignment with the source material, potentially due to vague or irrelevant contexts retrieved for this particular query. The response, while based on relevant retrieved contexts, deviates from the source material by emphasizing the user-friendliness of the Palm API through Maker Suite, which isn't explicitly mentioned in the retrieved contexts. The low factuality (0.00) and poor faithfulness (0.18) scores reflect this misalignment, indicating the generated answer includes unsupported details. Despite a high retrieval quality score (0.84), the system failed to focus on the specific context of the query, suggesting a gap in utilizing the most relevant information from the retrieved contexts. This indicates that the retrieval process effectively provides relevant context, but improvements in prompting the LLM are needed to prevent it from going off-topic when the available data is limited. Enhancing the prompting strategy could reduce over-generalization and help the model remain more context-aware.

6 Personal Contributions

Syed Ali Haider

The primary contribution to the project involved building an end-to-end pipeline to preprocess raw data and process multimodal inputs, including video and voice modalities. This framework served as the backbone of the project and allowed for further development and analysis. Also contributed to the creation of diverse encoders for evaluating the performance of different models in the system. Another significant contribution was developing the evaluation framework that allowed the project to be evaluated using various metrics. This framework was essential for measuring the effectiveness of the system. Contributed to the research and development of KR and KT methods, which played a crucial role in ensuring the scalability and reliability of the system.

Harsh Tambi

A key contribution included building a Retrieval-Augmented Generation (RAG) pipeline using ChromaDB for diverse content retrieval, adding a multimodal input pipeline (with Assembly AI and gpt-4o mini), and also adding a feature that shows users precise YouTube timestamps, helping them quickly locate specific sections in video lectures for deeper learning. The work in advanced RAG methods and prompt engineering ensured system accuracy, scalability, and reliability. These efforts resulted in a scalable and interactive chatbot tailored to diverse learning styles, offering accurate, context-aware responses and bridging gaps in traditional educational tools.

7 Conclusion

In this project, a multimodal RAG-based context-aware chatbot was engineered and the performance of the RAG system evaluated, focusing on retrieval quality, factuality, and faithfulness. Our findings indicate that, while the retrieval mechanism effectively identifies relevant context, the response generation sometimes deviates from the expected alignment, particularly in cases with limited data. This was reflected in the lower factuality and faithfulness scores, suggesting overgeneralization and gaps in context-awareness when context is scarce.

Future Work

To further enhance the system's performance, we propose the following future work. Explore more sophisticated RAG models and hybrid architectures to improve both the retrieval and generation quality. Refine prompting strategies to ensure better context retention and reduce over-generalization, especially when the data are scarce. Develop dynamic systems that select different LLMs based on query type. For example, using a math-focused LLM for math queries and a language-oriented model for textual queries, with reinforcement learning to optimize model selection. By addressing these areas, researchers can aim to further refine the ability of the RAG system to generate responses that are contextually relevant, factually accurate, and tailored to specific query types. These improvements will improve the flexibility and robustness of the system for various applications.

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Appendix

You can find our repository here to access the code